Software Tester (QA) – Home Assignment

Target: staging.ablespace.io

Section 1: Application Access

- **Target Application:** `staging.ablespace.io`
- **Account Setup:** Account created manually via standard signup flow.

Section 2: Exploratory Testing Scenarios

1. **Email Verification Banner Persistence:** Checked if the "Unconfirmed email" warning banner disappears after triggering "Resend Verification Email" or logging out/in. *(Result: Banner remains visible indefinitely until full confirmation page load.)*
2. **Student Creation with Boundary Inputs:** Added a student with a 255-character name string and special characters (`<script>`, `#@$%`). *(Result: Input accepted without validation warning, leading to text truncation on list view.)*
3. **Caseload Search Filter Persistence:** Applied a search filter on the Caseload page, navigated to Calendar, and returned via back navigation. *(Result: Search filter state was reset to default.)*
4. **Data Collection Form Empty Fields:** Attempted submitting "Take Data" entries with zero numerical entries and blank session notes. *(Result: Form submitted successfully without requiring minimal valid metrics.)*
5. **Page Refresh During Active Data Session:** Refreshed browser during an ongoing live data collection session for Demo Student1. *(Result: In-progress data session cleared with no auto-save or restore option.)*
6. **Concurrent Login Handling:** Logged into the same account simultaneously from two different browsers (Chrome and Firefox). *(Result: Session maintained on both without forced sign-out or session conflict notification.)*
7. **Service Time Categorization Toggle:** Toggled service time categorizations rapidly back and forth between 30 and 60 minutes. *(Result: Delayed UI state update causing incorrect category label rendering.)*
8. **Invalid Date Formatting in IEP Due Field:** Manually typed non-date values (e.g., `99/99/2026`) into the IEP Due date picker field. *(Result: System accepted invalid string and displayed broken date in table.)*
9. **Accommodation Tracker Empty State:** Accessed Accommodations section for a student with no assigned accommodations. *(Result: Screen displays infinite loading spinner instead of a clean empty-state message.)*
10. **Collaborator Permission Limits:** Tested adding a collaborator without administrative access. *(Result: User could still edit student service time settings.)*
11. **Google Sign-In Popup Dismissal:** Clicked "Continue with Google" on the login page, then closed the popup without selecting an account. *(Result: A raw technical Firebase error toast ("Firebase: Error (auth/popup-closed-by-user)") was shown to the user.)*
12. **Clever Sign-In Flow:** Clicked "Continue with Clever" on the login page. *(Result: Error "We couldn't navigate you to that URI" appeared; clicking "Try again" then worked.)*

13. **Browser Back Navigation After Clever Badge Login:** After the Clever flow, clicked the badge and then used the browser Back button. *(Result: Back first lands on the intermediate school page; a second Back press is needed to return to the home page.)*
14. **Temporary / Disposable Email Signup:** Attempted signup with a temporary (disposable) email address. *(Result: No check exists; temporary email domains are not blocked or flagged.)*
15. **Whitespace in Email Field:** Entered an email with leading or trailing spaces during signup/login. *(Result: Validation error shown, but the email looks correct to the user because trailing spaces are not visible. Input is not auto-trimmed.)*
16. **Role Dropdown Hover Behavior:** Hovered over the role dropdown on the signup form. *(Result: Cursor does not change to a pointer.)*
17. **Password Field Usability & Recovery:** Reviewed the password field on signup/login. *(Result: No show/hide password toggle and no "Forgot password" option.)*
18. **Feature Selection Layout (Onboarding):** Reached the "select features you'll use" step during onboarding. *(Result: The Skip button overlaps the info text.)*
19. **Take Data Counter Behavior:** Opened "Take Data" on a session and used the + (increase) and Undo controls on the counter. *(Result: Counter initially shows "-", switches to 0 after the first click, and Undo never goes below 0, which is confusing.)*

Section 3: Detailed Test Cases

Chosen Workflow: Create Student → Enter Details → Save → Search/View Student → Edit Student

Rationale: This is the core lifecycle workflow for educators and specialists managing caseloads on AbleSpace.

TC-01: Verify Successful Creation of New Student with Valid Data

Test Case ID	TC-01
Title	Create new student with all required fields valid
Preconditions	User is logged in and navigated to `staging.ablespace.io/caseload`
Test Data	First Name: `Alex`, Last Name: `Testing`, IEP Due: `10/30/2026`, Eval Due: `10/25/2026`, Service Time: `30`
Steps	1. Click **"Add Student"** button on the top right. 2. Fill in First Name, Last Name, IEP Due, Eval Due, and Service Time. 3. Click **"Save"**.
Expected Result	Student record is saved, modal closes, and `Alex Testing` appears on the Caseload table.
Actual Result	Student created successfully and listed on Caseload table.
Status	**Pass**

TC-02: Verify Validation Error When Creating Student with Blank Required Fields

Test Case ID	TC-02
Title	Attempt to create student with empty First Name and Last Name
Preconditions	User is logged in and on the `Add Student` modal
Test Data	First Name: `[Empty]`, Last Name: `[Empty]`
Steps	1. Open **"Add Student"** modal. 2. Leave First Name and Last Name blank. 3. Click **"Save"**.
Expected Result	System prevents submission and highlights required fields with clear error messages.
Actual Result	Form submission prevented; inline red validation errors displayed.
Status	**Pass**

TC-03: Verify Search Functionality Filters Caseload Correctly

Test Case ID	TC-03
Title	Search student by exact First Name
Preconditions	Caseload contains at least two students: `Demo Student1` and `Demo Student2`
Test Data	Search Term: `Student1`
Steps	1. Navigate to `staging.ablespace.io/caseload`. 2. Click on the **"Search students..."** input field. 3. Type `Student1`.
Expected Result	Caseload table filters dynamically to show only `Demo Student1`.
Actual Result	Only `Demo Student1` is displayed; `Demo Student2` is hidden.
Status	**Pass**

TC-04: Verify Editing Existing Student Details Updates Table Correctly

Test Case ID	TC-04
Title	Update IEP Due date and Service Time for existing student
Preconditions	`Demo Student1` exists on Caseload list
Test Data	Updated IEP Due: `12/01/2026`, Updated Service Time: `45`
Steps	1. Click the action menu (⋮) next to `Demo Student1`.

	2. Select **"Edit Student"**. 3. Change IEP Due to `12/01/2026` and Service Time to `45`. 4. Click **"Save"**.
Expected Result	Student details update in the database and display updated values on the table.
Actual Result	Details updated successfully.
Status	**Pass**

TC-05: Verify Handling of Special Characters and XSS in Student Name

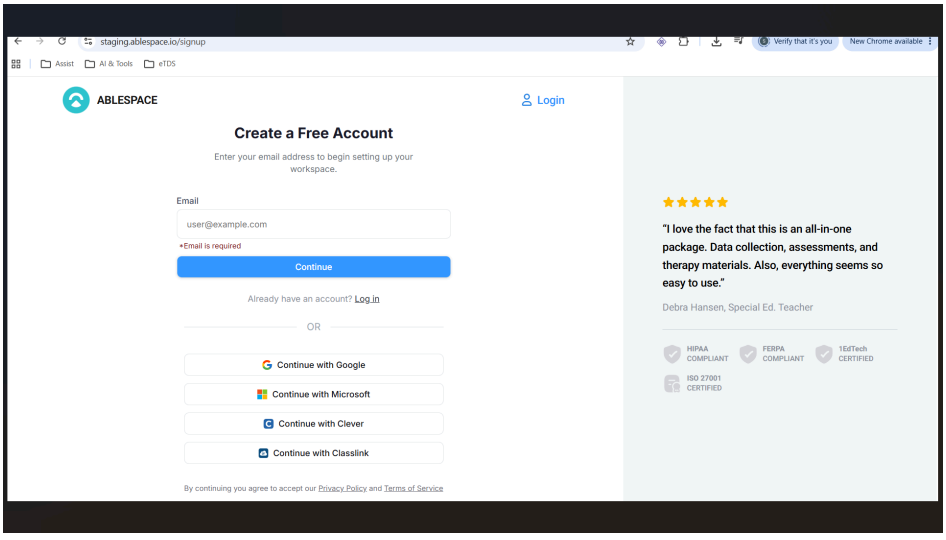
Test Case ID	TC-05
Title	Enter HTML/script injection payload into Student First Name field
Preconditions	User on "Add Student" modal
Test Data	First Name: `<script>alert('test')</script>`
Steps	1. Paste `<script>alert('test')</script>` into First Name field. 2. Enter valid details for remaining fields. 3. Click **"Save"**.
Expected Result	Input is sanitized or safely escaped as plain text without executing scripts or breaking rendering.
Actual Result	Input saved as sanitized plain text; no script executed.
Status	**Pass**

TC-06: Verify Deleting/Archiving Student from Caseload

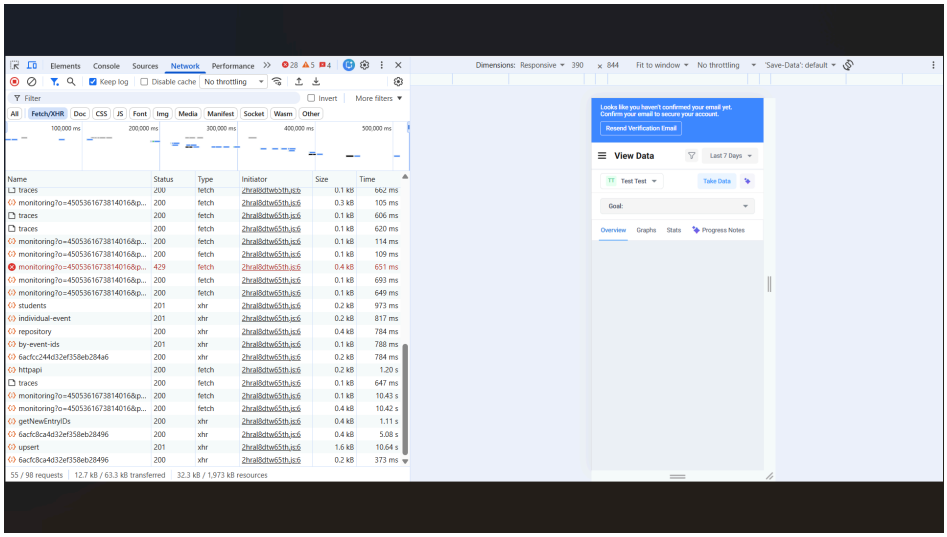
Test Case ID	TC-06
Title	Delete student with confirmation prompt
Preconditions	At least one test student exists in Caseload
Test Data	Target: `Demo Student2`
Steps	1. Click action menu (⋮) for `Demo Student2`. 2. Select **"Delete"** / **"Remove"**. 3. Confirm the modal action.
Expected Result	Confirmation dialog appears; upon confirming, student is removed from active Caseload.
Actual Result	Student removed successfully.
Status	**Pass**

Section 4: Bug Reports

BUG-01: Unwanted triggering of the email validation error message

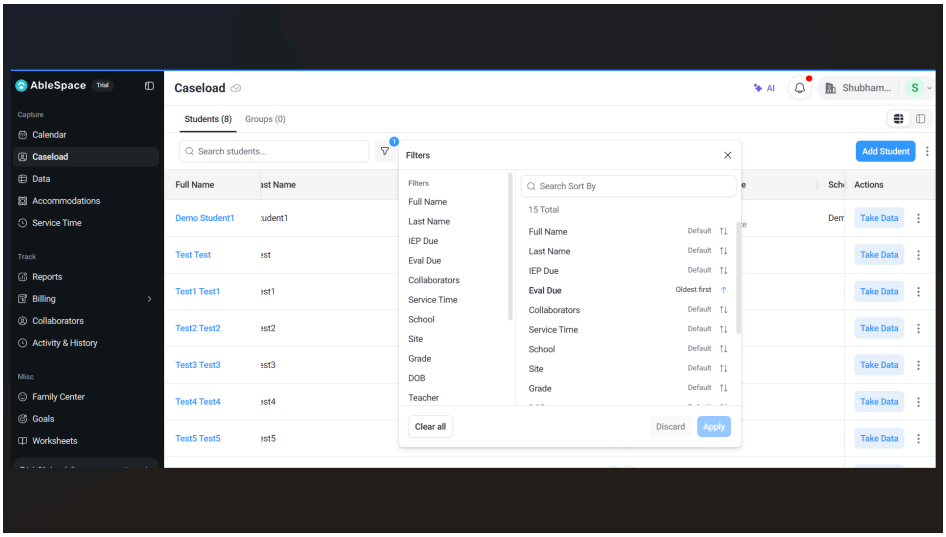
Bug ID	BUG-01
Title	Unwanted triggering of the email validation error message
Severity	Medium
Priority	Medium
Environment	https://staging.ablespace.io/caseload
Browser / Device	Chrome 129 on Windows 11
Reproducibility	Always
Steps to Reproduce	1. Navigate to the Login page. 2. Click inside the Email field. 3. Click anywhere outside the Email field without entering any text. 4. Observe whether the email validation error message is displayed.
Expected Result	The application should process resend requests appropriately, prevent repeated clicks through a temporary cooldown and rate limiting, display a loading indicator, and provide clear success or error notifications. .
Actual Result	No loading indicator, cooldown, rate-limit feedback, or confirmation message is displayed, and the button remains enabled for repeated clicks without clear feedback.
Evidence	

BUG-02: Action Menu Overflow Cut Off on Lower Viewports

Bug ID	BUG-02
Title	Caseload table action dropdown (⋮) gets clipped below viewport on short screens
Severity	Medium
Priority	Low
Environment	https://staging.ablespace.io/caseload
Browser / Device	Chrome (390 x 844 Mobile Emulation)
Reproducibility	Always
Steps to Reproduce	1. Open Caseload table on a mobile browser or viewport size 390x844. 2. Click the three-dot menu (⋮) for the last student row.
Expected Result	Dropdown menu auto-positions upward or inside scroll container so all options are accessible.
Actual Result	Options are cut off offscreen without vertical scroll access.
Evidence	

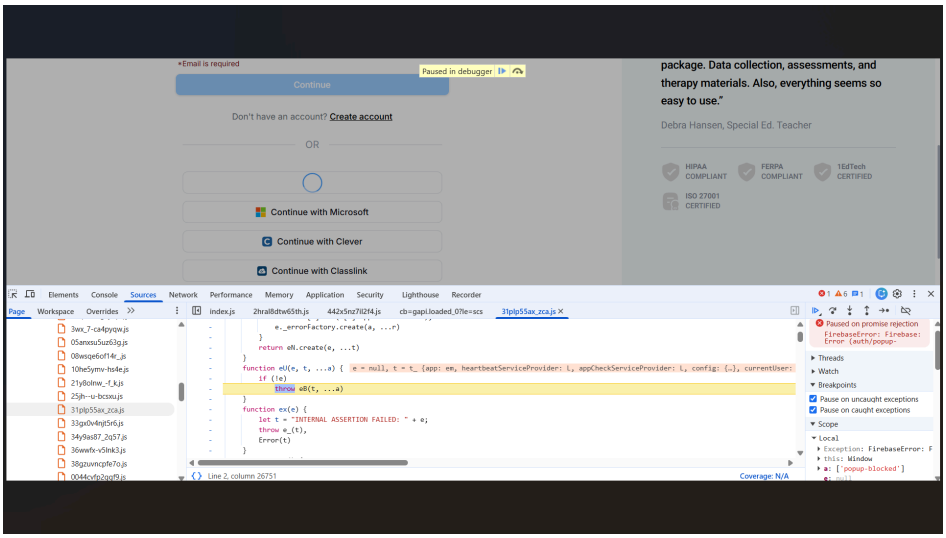
BUG-03: Filter Badge Counter Shows Active Filter When Filter Search Field Is Cleared

Bug ID	BUG-03
Title	Filter indicator badge shows `(1)` even after clearing input filter text
Severity	Low
Priority	Low

Environment	https://staging.ablespace.io/caseload
Browser / Device	Edge / Chrome Windows 11
Reproducibility	Always
Steps to Reproduce	1. Click filter funnel icon next to search input on Caseload page. 2. Type search text; observe badge shows blue indicator `(1)`. 3. Clear text using backspace or clear button.
Expected Result	Filter badge count resets to zero / disappears.
Actual Result	Badge remains active with blue `(1)` count.
Evidence	

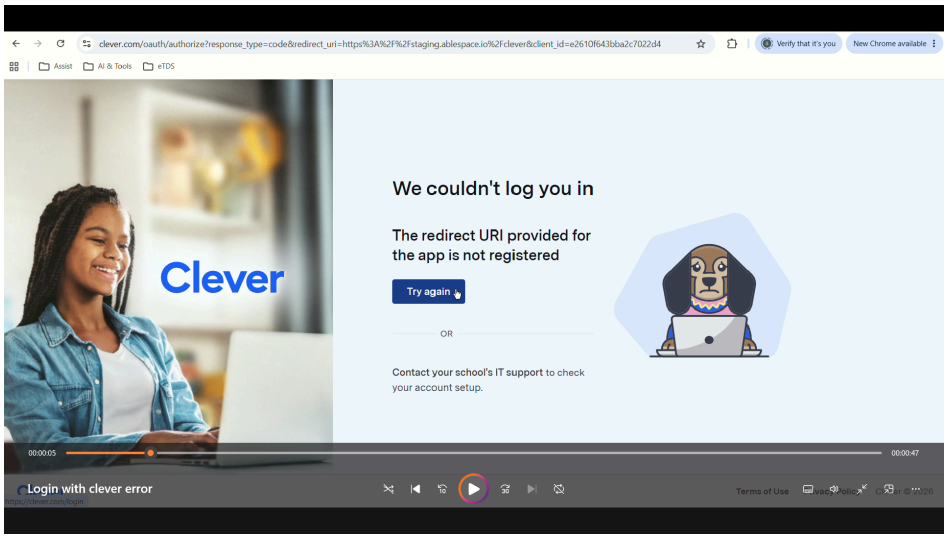
BUG-04: Raw Firebase Error Toast Shown When Google Sign-In Popup Is Closed

Bug ID	BUG-04
Title	Closing the "Continue with Google" popup displays a technical Firebase error message to the user
Severity	Low
Priority	Medium
Environment	https://staging.ablespace.io/login
Browser / Device	Desktop browser
Reproducibility	Always
Steps to Reproduce	1. Open the login page. 2. Click **"Continue with Google"**. 3. Close the Google popup without selecting an account.

Expected Result	No error is shown, or a friendly message appears (e.g., "Sign-in was cancelled. Please try again.").
Actual Result	A technical toast is displayed: "Firebase: Error (auth/popup-closed-by-user)". Internal error codes are exposed to the user.
Evidence	

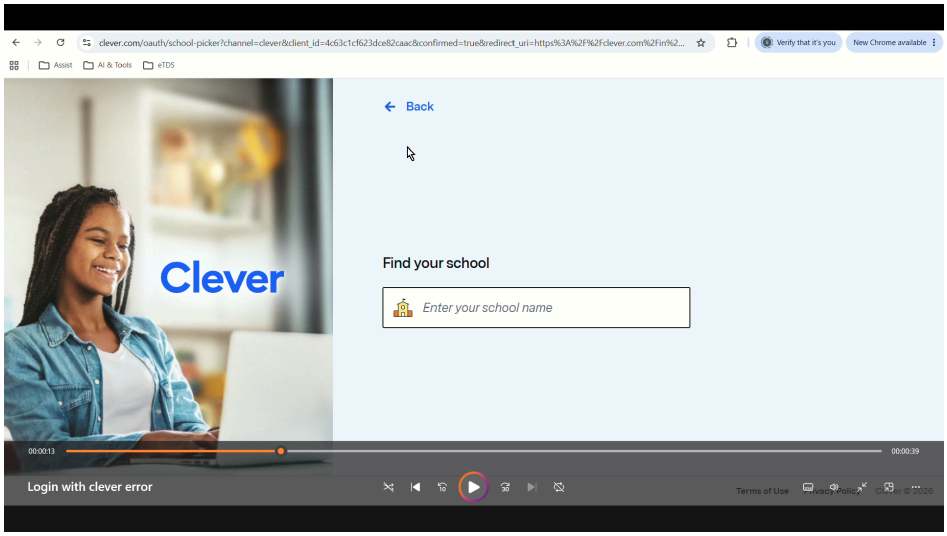
BUG-05: Clever Sign-In Fails on First Attempt with "Couldn't Navigate You to That URI" Error

Bug ID	BUG-05
Title	Continue with Clever shows a navigation error; works only after clicking "Try again"
Severity	Medium
Priority	High
Environment	https://staging.ablespace.io/login
Browser / Device	Desktop browser
Reproducibility	Always
Steps to Reproduce	1. Open the login page. 2. Click **"Continue with Clever"**. 3. Observe the error message "We couldn't navigate you to that URI". 4. Click **"Try again"**.
Expected Result	Clicking "Continue with Clever" takes the user directly to the Clever sign-in flow on the first attempt.

Actual Result	The error appears on the first attempt. The flow works only after clicking "Try again", adding friction and suggesting a redirect URI or configuration problem.
Evidence	

BUG-06: Browser Back Button Requires Two Presses After Clever Badge Login

Bug ID	BUG-06
Title	Back from the Clever badge flow lands on an intermediate school page before returning to the home page
Severity	Low
Priority	Medium
Environment	https://staging.ablespace.io/login
Browser / Device	Desktop browser
Reproducibility	Always
Steps to Reproduce	1. Open the login page and click **"Continue with Clever"** (use "Try again" if the URI error appears, see BUG-05). 2. Click the Clever badge. 3. Click the browser **Back** button.
Expected Result	A single Back press returns the user to the previous page (home page).
Actual Result	Back first navigates to the school page. The user must press Back a second time to reach the home page, indicating redundant history entries from the redirect chain.

Evidence	

BUG-07: No Validation Against Temporary / Disposable Email Addresses at Signup

Bug ID	BUG-07
Title	Signup accepts emails from temporary-mail providers
Severity	Medium
Priority	Medium
Environment	https://staging.ablespace.io/login
Browser / Device	Desktop browser
Reproducibility	Always
Steps to Reproduce	1. Open the signup page. 2. Enter an email address from a disposable/temporary mail service. 3. Complete the remaining fields and submit.
Expected Result	System blocks or warns about disposable email domains to protect data quality and reduce fake or abusive account creation.
Actual Result	No check exists; the account is created with a temporary email address.
Evidence	N/A

BUG-08: Leading/Trailing Spaces in Email Field Are Not Auto-Trimmed and Cause a Confusing Error

Bug ID	BUG-08

Title	Email with a trailing or leading space is rejected, but the error is unclear because the space is not visible
Severity	Low
Priority	Medium
Environment	https://staging.ablespace.io/login
Browser / Device	Desktop browser
Reproducibility	Always
Steps to Reproduce	1. Open the signup or login page. 2. Enter a valid email followed by a trailing space (or preceded by a leading space). 3. Submit the form.
Expected Result	Spaces before and after the email are trimmed automatically and the form proceeds normally.
Actual Result	An error is displayed even though the email looks correct. Users cannot easily see a trailing space and cannot tell why the email is rejected.
Evidence	N/A

BUG-09: Role Dropdown Does Not Show Pointer Cursor on Hover (Signup)

Bug ID	BUG-09
Title	Hovering the role dropdown on the signup form keeps the default cursor
Severity	Low
Priority	Low
Environment	https://staging.ablespace.io/login
Browser / Device	Desktop browser
Reproducibility	Always
Steps to Reproduce	1. Open the signup page. 2. Hover the mouse over the **Role** dropdown.
Expected Result	Cursor changes to a pointer, signalling the element is clickable.
Actual Result	Cursor stays as the default arrow, inconsistent with other interactive elements.
Evidence	N/A

BUG-10: Password Field Has No Show/Hide Visibility Toggle

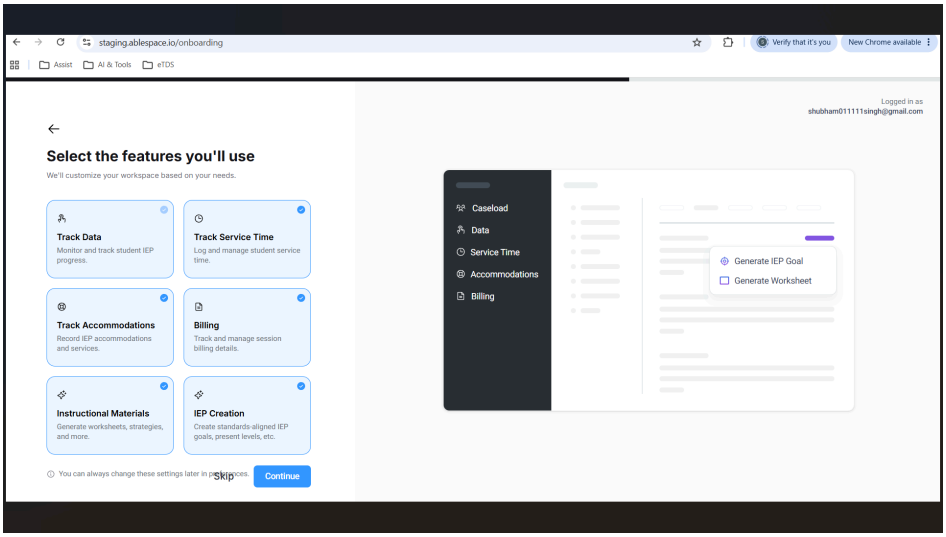
Bug ID	BUG-10
Title	Users cannot reveal the password they are typing
Severity	Low
Priority	Low
Environment	https://staging.ablespace.io/login
Browser / Device	Desktop browser
Reproducibility	Always
Steps to Reproduce	1. Open the signup or login page. 2. Click into the password field and type a password.
Expected Result	An eye icon or similar control lets the user toggle password visibility.
Actual Result	No toggle is available, increasing the chance of typos, especially on mobile.
Evidence	N/A

BUG-11: No "Forgot Password" Option on the Login Page

Bug ID	BUG-11
Title	Users who forget their password have no recovery path
Severity	High
Priority	High
Environment	https://staging.ablespace.io/login
Browser / Device	Desktop browser
Reproducibility	Always
Steps to Reproduce	1. Open the login page. 2. Look for a password recovery or reset link near the password field and sign-in button.
Expected Result	A "Forgot password?" link starts a password reset flow via email.
Actual Result	No password recovery option exists. Users who forget their password are locked out of the account.
Evidence	N/A

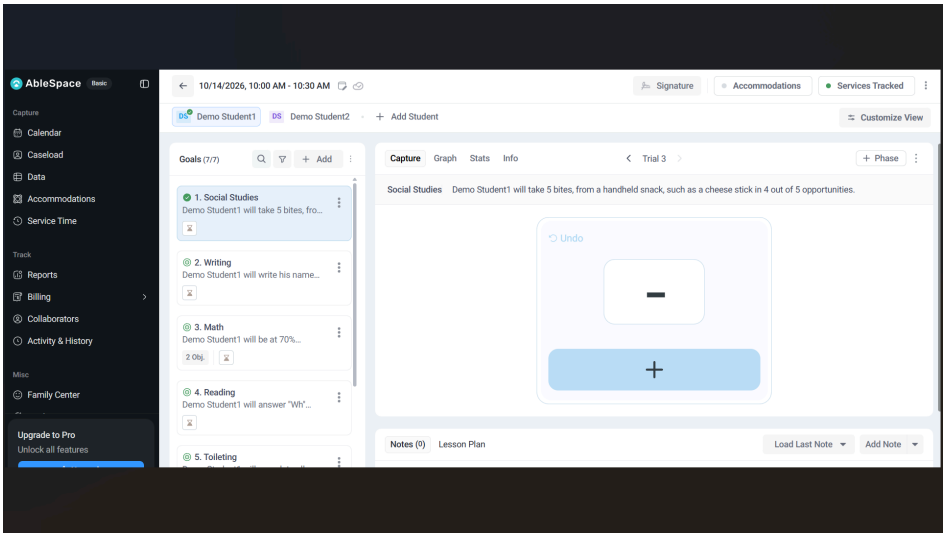
BUG-12: Skip Button Overlaps Info Text on the Feature Selection Step

Bug ID	BUG-12

Title	During onboarding, the Skip button covers the info text on the "select features you'll use" screen
Severity	Medium
Priority	Medium
Environment	staging.ablespace.io (onboarding flow after signup)
Browser / Device	Desktop browser
Reproducibility	Always
Steps to Reproduce	1. Sign up with a new account and continue through onboarding. 2. Reach the step where you select the features you will use. 3. Observe the position of the **Skip** button relative to the info text.
Expected Result	Skip button and info text are laid out separately with no overlap, and all text is readable.
Actual Result	The Skip button overlaps the info text, making part of the text hard to read.
Evidence	

BUG-13: Take Data Counter Shows "-" Initially and Cannot Be Undone Below 0

Bug ID	BUG-13
Title	Counter starts at "-", jumps to 0 on the first click, and Undo never goes below 0
Severity	Medium
Priority	Medium
Environment	staging.ablespace.io (Take Data on a session)
Browser / Device	Desktop browser

Reproducibility	Always
Steps to Reproduce	1. Open any student session and click **"Take Data"**. 2. Observe the counter: it initially shows "-". 3. Click the **+** button once; observe the counter value. 4. Continue clicking **+**, then click **Undo**.
Expected Result	The initial state is clear and consistent (e.g., 0 initially, and the first + click shows 1). Undo steps the count back down predictably, and the user can see how to return to the initial state.
Actual Result	The counter initially shows "-"; after the first click it shows 0 and then increments. Undo stops at 0 and never goes lower or back to "-", which is confusing about whether the first click was recorded.
Evidence	

Section 5: Cross-Browser & Responsive Testing Matrix

Area / Feature	Chrome (Desktop)	Firefox / Edge (Desktop)	Mobile (390 × 844)	iPad (Tablet)	Notes
Signup and Login	**Pass**	**Pass**	**Pass**	**Pass**	Google/Firebase popups require explicit popup permission on Chrome. See also BUG-04, BUG-05, BUG-10, BUG-11.
Main Workflow (Caseload)	**Pass**	**Pass**	**Fail**	**Pass**	Action dropdown cut off at bottom row on mobile view (BUG-02).
Forms (Add/Edit Student)	**Pass**	**Pass**	**Pass**	**Pass**	Responsive layout wraps inputs cleanly.

Area / Feature	Chrome (Desktop)	Firefox / Edge (Desktop)	Mobile (390 × 844)	iPad (Tablet)	Notes
Navigation & Sidebar	**Pass**	**Pass**	**Pass**	**Pass**	Collapsible sidebar functions smoothly on all sizes.

Section 6: Test Summary Report

1. Testing Performed

Executed exploratory, functional, visual, and cross-browser responsiveness tests on `staging.ablespace.io`. Platforms covered include Google Chrome Desktop (Windows 11), Microsoft Edge, Mobile viewport (390x844), and iPad Mini/Pro emulations. Testing covered the core caseload workflow as well as authentication (Google and Clever sign-in, password handling), onboarding, and the Take Data experience.

2. Issues Found

Documented **13 bugs** (BUG-01 to BUG-13) and 2 usability minor observations. Three bugs relate to the core caseload workflow (BUG-01 to BUG-03); ten relate to authentication, onboarding, and data collection (BUG-04 to BUG-13).

- **Fix Priority:** BUG-01 (Resend Verification Email lack of feedback/rate-limiting) should be addressed first. Without feedback or rate limiting, users can cause server spam or assume email functionality is non-responsive.
- **Also high priority:** BUG-11 (no Forgot Password option) and BUG-05 (Clever sign-in fails on first attempt) directly block or hinder account access and should be fixed alongside BUG-01.
- **Security / data quality:** BUG-07 (no disposable email check) should be reviewed to reduce fake account creation.

3. Areas Not Tested

- **Automated bulk account creation & security penetration/load testing:** Excluded per explicit assignment rules.
- **Billing & Payment Gateways:** Skipped to prevent initiating real/test transaction workflows without sandbox configurations.

4. What I Would Test Next (Given 1–2 Extra Hours)

- **Data Export & Reports:** Test PDF/CSV generation accuracy for student progress and service time tracking.
- **Session Timeout & Security Tokens:** Verify session expiration behavior after prolonged inactivity.

Section 7: Optional Playwright Automation (Bonus)

Playwright Test Code (tests/caseload.spec.js)

```
// @ts-check
const { test, expect } = require('@playwright/test');

test.describe('AbleSpace Caseload Management Flow', () => {

  test.beforeEach(async ({ page }) => {
    // Read credentials from environment variables
    const email = process.env.ABLESPACE_EMAIL;
    const password = process.env.ABLESPACE_PASSWORD;

    if (!email || !password) {
      throw new Error('ABLESPACE_EMAIL and ABLESPACE_PASSWORD environment variables must be set.');
    }

    // Login flow
    await page.goto('https://staging.ablespace.io/login');
    await page.getByLabel(/email/i).fill(email);
    await page.getByLabel(/password/i).fill(password);
    await page.getByRole('button', { name: /sign in|log in/i }).click();

    // Verify successful redirection to caseload
    await expect(page).toHaveURL(/.*caseload/);
  });

  test('Should search for a student and display matching results', async ({ page }) => {
    const searchInput = page.getByPlaceholder(/search students/i);

    // Ensure search bar is visible
    await expect(searchInput).toBeVisible();

    // Search for student
    await searchInput.fill('Demo Student1');

    // Assert filtered table results
    const studentRow = page.getByRole('cell', { name: 'Demo Student1' });
    await expect(studentRow).toBeVisible();
  });
});
```

README.md for Playwright Suite

```
# AbleSpace Playwright Automation Suite

This package contains automated E2E tests for AbleSpace staging environment using Playwright.

## Prerequisites
- Node.js (v16 or higher)
- npm

## Setup Instructions

1. Install Dependencies:
   npm install
   npx playwright install

2. Configure Environment Variables: Create a .env file in the root directory
   (added to .gitignore):
   ABLESPACE_EMAIL=your_test_email@example
   ABLESPACE_PASSWORD=your_test_password

3. Run the tests:
   npx playwright test
```